

Queue Implementation using Array in C

Source Code

```
#include <stdio.h>
#include <stdbool.h>
#include <stdlib.h>

#define MAX_SIZE 10

int queue[MAX_SIZE];
int front = -1;
int rear = -1;

bool isEmpty();
bool isFull();
void enqueue();
int dequeue();
void traverse();

int main() {
    int choice;
    int item;

    while(1) {
        printf("\n===== QUEUE MENU =====\n");
        printf("\n1. Enqueue\n");
        printf("\n2. Dequeue\n");
        printf("\n3. Check if Full\n");
        printf("\n4. Check if Empty\n");
        printf("\n5. Traverse\n");
        printf("\n6. Exit\n");
        printf("\nEnter your choice: ");
        scanf("%d", &choice);

        switch(choice) {
            case 1:
                enqueue();
                break;
            case 2:
```

```

        item = dequeue();
        break;
    case 3:
        if(isFull())
            printf("\nQueue is FULL!\n");
        else
            printf("\nQueue is NOT full.\n");
        break;
    case 4:
        if(isEmpty())
            printf("\nQueue is EMPTY!\n");
        else
            printf("\nQueue is NOT empty.\n");
        break;
    case 5:
        traverse();
        break;
    case 6:
        printf("\nExiting...\n");
        exit(0);
    default:
        printf("\nInvalid choice! Try again.\n");
    }
}

return 0;
}

bool isEmpty() {
    if(rear == front || rear == -1){
        return true;
    }
    return false;
}

bool isFull() {
    if(rear == MAX_SIZE-1){
        return true;
    }
    return false;
}

```

```

void enqueue() {
    if(isFull()) {
        printf("\nQueue Overflow\n");
    } else {
        int ele;
        printf("\nEnter element to enqueue: ");
        scanf("%d",&ele);
        rear++;
        queue[rear] = ele;
        printf("\nElement enqueued..");
    }
}

int dequeue() {
    if(isEmpty()) {
        printf("\nQueue Underflow\n");
    } else {
        front++;
        return queue[front];
    }
}

void traverse() {
    if(isEmpty()) {
        printf("\nQueue is empty.\n");
    } else {
        printf("\nQueue elements:\n");
        for(int i = front+1; i <= rear; i++){
            printf("%d\t",queue[i]);
        }
    }
}

```

Output :

```
===== QUEUE MENU =====
```

1. Enqueue
2. Dequeue
3. Check if Full
4. Check if Empty
5. Traverse
6. Exit

Enter your choice: 1

Enter element to enqueue: 10

Element enqueued..

```
===== QUEUE MENU =====
```

1. Enqueue
2. Dequeue
3. Check if Full
4. Check if Empty

5. Traverse

6. Exit

Enter your choice: 1

Enter element to enqueue: 20

Element enqueued..

```
===== QUEUE MENU =====
```

1. Enqueue
2. Dequeue
3. Check if Full
4. Check if Empty
5. Traverse
6. Exit

Enter your choice: 5

Queue elements:

10 20

```
===== QUEUE MENU =====
```

```
1. Enqueue
2. Dequeue
3. Check if Full
4. Check if Empty
5. Traverse
6. Exit

Enter your choice: 3

Queue is NOT full.

===== QUEUE MENU =====

1. Enqueue
2. Dequeue
3. Check if Full
4. Check if Empty
5. Traverse
6. Exit
```

```
Enter your choice: 4

Queue is NOT empty.

===== QUEUE MENU =====

1. Enqueue
2. Dequeue
3. Check if Full
4. Check if Empty
5. Traverse
6. Exit

Enter your choice: 2

===== QUEUE MENU =====

1. Enqueue
2. Dequeue
3. Check if Full
```

```
4. Check if Empty
5. Traverse
6. Exit

Enter your choice: 5

Queue elements:
20

===== QUEUE MENU =====

1. Enqueue
2. Dequeue
3. Check if Full
4. Check if Empty
5. Traverse
6. Exit

Enter your choice: 2

===== QUEUE MENU =====

1. Enqueue
```